

EGYPTIAN MINISTRY MUSIC THEORY & ARAB/WESTERN MUSICOLOGY

Official Ministry Standardized Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (THANAWEYA AMMA)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Official Egyptian Ministry music curriculum covering acoustic wave fundamentals, staff reading in G and F clefs, classical harmony and cadences, 24-quarter-tone Maqam structures, Arab Iqa'at metrics, and historical pioneers.

CORE CURRICULUM HIGHLIGHTS:

- Acoustic physics of sound waves, frequency, wavelength, and overtone series
- Western staff notation fluency in Treble (G) and Bass (F) clefs with accidentals
- Classical voice leading, triad construction, dominant seventh V7, and harmonic cadences
- The 24 quarter-tone equal temperament system and analysis of the 8 principal Maqamat

APPROVED CHAPTER MODULES:

- Chapter 1: Acoustics, Sound Physics & Staff Notation	pp. 1 56 (4 core units)
- Chapter 2: Classical Harmony, Triads & Cadences	pp. 57 112 (4 core units)
- Chapter 3: Arab Music Theory & The 8 Maqamat	pp. 113 168 (4 core units)
- Chapter 4: Arab Rhythms, Organology & Renaissance Masters	pp. 169 224 (4 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Music Theory, Maqamat & Staff Notation Compendium (8p - 22.8 KB)
Full Ministerial Textbook: 224 Pages - 20.6 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: TH-MUS-G12
Legal Deposit: 2025/14892 - ISBN: 978-977-12-9649-13



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Official Egyptian Ministry music curriculum covering acoustic wave fundamentals, staff reading in G and F clefs, classical harmony and cadences, 24-quarter-tone Maqam structures, Arab Iqa'at metrics, and historical pioneers. Acoustic physics of sound waves, frequency, wavelength, and overtone series Western staff notation fluency in Treble (G) and Bass (F) clefs with accidentals Classical voice leading, triad construction, dominant seventh V7, and harmonic cadences The 24 quarter-tone equal temperament system and analysis of the 8 principal Maqamat

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
---	---	--

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Chapter 1: Acoustics, Sound Physics & Staff Notation	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Chapter 2: Classical Harmony, Triads & Cadences	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Chapter 3: Arab Music Theory & The 8 Maqamat	7 wks	15 pts	25.0%	4 core instructional units covering national standards and
Chapter 4: Arab Rhythms, Organology & Renaissance Masters	7 wks	15 pts	25.0%	4 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CHAPTER 1: ACOUSTICS, SOUND PHYSICS & STAFF NOTATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Chapter 1: Acoustics, Sound Physics & Staff Notation.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Chapter 1: Acoustics, Sound Physics & Staff Notation.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CHAPTER 2: CLASSICAL HARMONY, TRIADS & CADENCES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Chapter 2: Classical Harmony, Triads & Cadences.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Chapter 2: Classical Harmony, Triads & Cadences.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CHAPTER 3: ARAB MUSIC THEORY & THE 8 MAQAMAT

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Chapter 3: Arab Music Theory & The 8 Maqamat.
- [+] Synthesize and apply analytical frameworks across all 4 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Chapter 3: Arab Music Theory & The 8 Maqamat.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\left(\frac{1}{N}\right) * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

EGYPTIAN MINISTRY MUSIC THEORY & ARAB/WESTERN MUSICOLOGY	PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$	Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Equilibrium Balance Criterion: $\sum F_{net} = 0, \sum M_{pivot} = 0$	Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Continuity & Conservation Standard: $\text{div}(J) + \partial \rho / \partial t = 0$	Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Normalized Standard Score: $Z = (X - \mu) / \sigma$	Curricular Efficiency Ratio: $\eta = (\text{Output_actual} / \text{Input_standard}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION

NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: TH-MUS-G12

Curriculum Cycle: 2025/2026 Academic Year | National Edition

Official Portal: <https://moe.gov.eg/ar/elearning-content/>

All rights reserved (c) 2025 Ministry of Education & Technical Education - Egypt.

Unauthorized commercial reprinting or distribution without ministerial sanction is prohibited.